Robotics-for-Cacao Barter

Follow-up action item · Agroverse · 2026-08-06

Context. Four problems facing cacao farming: witches' broom disease, an aging farmer population with no labor to replace it, inconsistent roasting, and farm-succession risk that threatens agroforestry continuity. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, https://youtu.be/hg8qYrjyYCU). This follow-up turns that introduction into a concrete pilot.

1 · The four problems

Problem 1 — Witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*)

The single highest-leverage canopy disease in Bahia cacao: infected brooms must be pruned out **and removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees. Detection and pruning are manual, repetitive, and easy to miss.

Problem 2 — Aging farmer population + labor shortage

Cacao farmers are aging, and there is **no younger generation stepping in** — farm management and harvest depend on manual labor that is increasingly hard to find and afford. The physical work (climbing, pruning, pod-cutting, carrying) is heavy, seasonal, and unattractive to new workers. The farm's know-how is concentrated in aging owners who cannot scale it.

Problem 3 — Roasting consistency

Roasting — the step that defines the flavor profile — is a **sequential decision process with delayed reward**: the operator acts on the roast curve (heat, airflow, drum speed, time), and only *after* the batch do you learn whether the flavor profile landed. Today that is skilled-craft, tuned batch-by-batch by experienced tasters — **not scalable** and hard to reproduce consistently across batches, roasters, and operators.

Problem 4 — Farm succession risk (the existential one)

When a farmer is hurt, or simply **too old**, the farm gets sold. There is **no guarantee the new owner continues agroforestry** — they may clear the land, plant something else, or run it down. Every farm lost to succession is hectares of regenerated rainforest **permanently reversed**, directly threatening the **10,000-hectare regeneration goal**. The agroforestry system lives or dies with the individual owner — that is fragile.

2 · Working prototype reference — Frasky (IIT)

Video (Reuters): https://youtu.be/hg8qYrjyYCU — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).
- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.

• Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

Proof point: a permanent tree/vine crop's canopy work can be robotized. Cacao is the same category — this is the working prototype we point partners and farmers to.

Frasky capability (grape)	Cacao adaptation	Problems it addresses
Cluster monitoring via camera	Broom / pod detection via vision	P1 broom, P2 labor
Manipulate plants (robotic arm)	Broom pruning with cutter arm	P1 broom, P2 labor
Targeted spray treatments	Targeted biocontrol on brooms	P1 broom
—	Selective pod *harvest* (varied heights: trunk + branches)	P2 labor
—	Farm-management data (what to do, where, when)	P2 aging owners, P4 succession

3 · Roasting consistency as a reinforcement learning problem

The reframe — this is a reinforcement learning problem.

RL element	In the cacao roaster
State (observation)	Bean moisture, bean temperature / rate-of-rise, roast color, particle size distribution, volatile markers
Action	Heat input, airflow, drum speed, batch timing — the roast-curve decisions
Reward	Flavor-profile match (sensor + taste panel), batch-to-batch consistency, energy efficiency
Policy	A learned roast curve that reproduces a target flavor profile, batch after batch

The agent **learns the roast policy** over batches — each batch is a training episode, the reward comes at the end. That is exactly what makes it RL rather than a static recipe: the controller adapts to bean lot, moisture, and machine state to hit the flavor target.

The observation layer — particle / roast-state detection. The RL agent is only as good as what it can sense:

Detection approach	What it measures	Status
Machine vision (camera)	Bean color, surface browning, particle size distribution	■ proven in food lines
Hyperspectral / SWIR imaging	Moisture, roast degree, bean defects (slaty, moldy, under/over-fermented)	■ proven, conveyor-scalable

Detection approach	What it measures	Status
Portable NIR spectroscopy	Chemical markers of roast degree, in-line	■ proven, low-cost
Electronic nose + ANN	Volatile profile → predicted roast degree (94.4% accuracy reported)	■ proven

Why now / why us. Precedent exists (IMA Group's "AI Learning to Roast", model-based roast optimization) but is sold as expensive industrial software. A robotics + ML partner can build a **low-cost RL roast controller + particle detector** tuned to the beans and flavor profiles the Agroverse network actually runs — turning roast flavor control from craft into a **scalable, self-improving process**.

4 · The proposal (barter model)

Offer robotics + automation services to Bahia/Pará farmers **in exchange for cacao as payment**:

Role	Who
Robot design & build	Robotics design expert (Gianluca)
Farmer network & deployment	Agroverse — Bahia operations (Ilhéus)
Market for cacao received	Agroverse (US import live)

The four problems → four workstreams:

Problem	Robotic/AI response	Feasibility
P1 · Witches' broom	Broom detection + pruning robot (à la Frasky)	■ tractable now
P2 · Aging farmers / labor	Harvest assist + farm-management data (what/where/when)	■ tractable (harvest ■■ harder R&D)
P3 · Roast consistency	RL roast controller + particle detection	■ parallel, lower-field-risk track
P4 · Succession risk	Farm runs with **far less physical labor** → owner stays productive longer; farm-management data makes the operation **ownable/transferable**; the cacao-barter agreement can carry an **agroforestry-continuity clause** so value stays tied to the standing trees	■ structural, low-tech-risk

Why P4 matters most. P1–P3 improve the farm. P4 **keeps the farm from being lost**. Robotics + the barter agreement together can make an agroforestry cacao farm *more valuable as agroforestry* than as cleared land — the strongest possible protection for regenerated hectares.

Honest MVP scoping. Broom *detection + pruning* (or monitoring + targeted treatment) is tractable now. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable.

5 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (https://youtu.be/hg8qYrjyYCU) as the reference prototype; define MVP build, season timing, and cost basis; evaluate the roast-RL track (state sensors, reward signal, data capture per batch).
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, aging-owner relief, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL; succession protection (the farm stays productive and valuable as cacao agroforestry).
3. **2–3 test farms** in the Bahia (Ilhéus) network.
4. **Barter accounting** — value cacao received at market BRL/USD and record it into the supply ledger at fair market value; draft the agroforestry-continuity clause for the agreement.
5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved, roast batch-to-batch consistency, **farms retained in agroforestry across succession events**. No scaling on unmeasured evidence.

6 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility + roast-RL questions: observation sensors, reward definition, data per batch) and the PT-BR farmer pitch.
- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

7 · Tie-back to mission

Every broom pruned, every robot-assisted harvest, every aging farmer given a tool instead of a shovel — and every batch of cacao roasted to a reproducible flavor profile — keeps more trees producing and more **Amazon rainforest standing**. And when a farm passes to new hands, the robotics + cacao-market system makes **continuing agroforestry the profitable choice** — that is how we reach and hold **10,000 hectares regenerated**.

Draft v9 · pending approval before any outreach.